

A Secure Multimodal Retrieval Framework for Consumer Electronics Under AI-Driven Attacks

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Abstract—In the field of consumer electronics, multimodal retrieval systems are increasingly employed to support intuitive product search based on a combination of reference images and natural language queries. However, such systems are vulnerable to AI-driven attacks, including adversarial prompts and manipulated inputs that can mislead retrieval outcomes. To address this, we propose SHRC-Net, a secure and robust multimodal image retrieval framework that integrates Semantic Harmonization and Region Composition. The architecture comprises three core modules: a Semantic Harmonization Unit (SHU) that aligns textual semantics with fine-grained visual regions, an Adaptive Region Composer (ARC) that captures region-level dependencies under semantic guidance, and a Unified Training Objective (UTO) that incorporates alignment, attention regularization, and compositional classification losses. SHRC-Net builds target-aware representations that faithfully reflect user intent while enhancing resistance to adversarial manipulations. The framework is trained end-to-end without requiring additional annotations. Experiments on multiple benchmark datasets demonstrate that SHRC-Net achieves state-of-the-art retrieval performance, while also exhibiting strong robustness against ambiguous or malicious queries. These results underscore its potential for secure deployment in AI-powered consumer electronics systems.

Index Terms—Multimodal image retrieval, consumer electronics, semantic harmonization, region composition.

I. INTRODUCTION

IN RECENT years, the rapid development of consumer electronics has fueled an increasing demand for intelligent visual search systems that support natural and multimodal user interactions [1], [2], [3], [4]. Users often express their retrieval intent through a reference image along with a textual description indicating specific changes. For example, a user might search for “a black version of this phone” or “a laptop with a larger screen and the same keyboard layout.” These

interactions are common in product recommendation systems and e-commerce platforms, where consumer decision-making depends on small but important visual and functional variations [5], [6], [7], [8]. Compared with conventional image retrieval tasks, this setting poses unique challenges: consumer electronics have rigid, well-defined parts such as screens, lenses, and ports, whose configurations are central to product identity. Therefore, the ability to recognize and reason over fine-grained, component-level variations is crucial. Meanwhile, as multimodal systems become more prevalent in AI-integrated consumer devices, they also introduce potential vulnerabilities to adversarial manipulation. Malicious users or automated agents may craft queries, either textual or visual in form, that are intentionally designed to mislead retrieval models, trigger incorrect matches, or exploit model biases. In safety- or privacy-critical applications, such threats can result in compromised user trust or unauthorized access to product-level information. As a result, it has become increasingly important to build retrieval systems that are not only accurate, but also secure and robust against adversarial inputs.

This task intersects with key areas in computer vision and natural language processing, including multimodal learning, cross-modal retrieval, image-text matching, and visual-language grounding. While large-scale pretrained models such as CLIP [9], ALIGN [10], and Florence [11] have demonstrated strong performance in general-purpose multimodal tasks, they often lack the region-level awareness and structural reasoning capacity needed for product-specific retrieval. In addition, object-level modeling has become increasingly important in fields such as object detection [12], vision-language pretraining [13], and multimodal transformers [14]. Approaches that emphasize region representations [15] and multimodal fusion [16] offer a promising foundation, but still face challenges when applied to structured and functionally rich categories like consumer electronics. Furthermore, most existing approaches do not explicitly address robustness under adversarial or ambiguous queries, which are increasingly relevant in real-world deployments.

Despite progress in multimodal learning, existing methods fall short when it comes to fine-grained image retrieval in product-centered domains. A common limitation is the use of global image-text embeddings, where the entire image and query are projected into a shared space [17], often ignoring localized relationships. This leads to semantic mismatches, especially when user intent focuses on only a specific part of the image. Some models introduce attention mechanisms

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to associate text with image regions [18], but suffer from issues such as attention collapse or weak supervision, which degrade alignment quality. Transformer-based architectures like ViLT [19] and LXMERT [20] offer end-to-end multimodal encoding, but typically do not account for region composition or structured changes. Scene graph and region-based grounding methods [21] provide explicit object-level reasoning, but rely on expensive annotations and are not easily scalable. Image editing or generation approaches such as StyleCLIP [22] or image synthesis-based retrieval [23] attempt to model visual transformations, but require generative capabilities and are less suited to retrieval settings. Moreover, recent fusion models like Co-Attention networks [24] and cross-modal transformers [25] do not explicitly capture the spatial structure of components, which is essential for reasoning over physical changes. In terms of optimization, most approaches rely on contrastive or ranking losses [26], which focus on global matching and lack explicit structural supervision. Although some recent efforts incorporate regularization or auxiliary tasks [27], they are often tailored to specific domains or datasets, and fail to generalize to the complexity of real-world product configurations [28].

To address these challenges, we propose SHRC-Net, a multimodal image retrieval framework tailored for the consumer electronics domain that integrates Semantic Harmonization and Region Composition. SHRC-Net is designed to model fine-grained cross-modal relationships and structural transformations implied by user queries, while maintaining robustness against adversarial manipulations. The overall architecture consists of three key components. First, the Semantic Harmonization Unit (SHU) aligns region-level visual features from the reference image with semantic cues from the query text. This alignment is achieved through contrastive learning in a shared embedding space, enabling localized grounding of semantic concepts. Second, the Adaptive Region Composer (ARC) refines the aligned features using enhanced self-attention and cross-modal semantic guidance. It captures spatial dependencies and generates a composite representation that reflects the structural intent behind the query. Third, the Unified Training Objective (UTO) consists of three complementary loss functions: a semantic alignment loss that enforces consistency between image regions and text segments; an attention regularization loss that encourages stable and diverse attention distributions; and a composition-based classification loss that enhances feature discriminability with respect to the target image. These modules work in concert to enable the network to interpret user intent at both semantic and structural levels, compose expressive features, and retrieve images that reflect fine-grained changes in appearance, components, or layout. SHRC-Net requires no additional annotations and is trained end-to-end using standard triplet-based datasets. Extensive experiments validate its effectiveness across a range of consumer electronics retrieval scenarios, including color variant identification, component replacement, and configuration-aware recommendation. Notably, we further demonstrate that SHRC-Net maintains retrieval accuracy even under adversarial input conditions, underscoring its suitability for secure deployment in AI-driven consumer environments.

The contributions are summarized as follows: (1) We propose SHRC-Net, a novel multimodal retrieval framework specifically designed for consumer electronics, which integrates Semantic Harmonization and Region Composition to model fine-grained cross-modal relationships and structural transformations. (2) The framework introduces a Semantic Harmonization Unit (SHU) to achieve robust region-level alignment between visual and textual modalities through bidirectional contrastive learning and semantic gating, and an Adaptive Region Composer (ARC) to capture spatial dependencies and semantically guided component variations. (3) A Unified Training Objective (UTO) combining alignment, attention regularization, and classification losses is developed to ensure stable optimization and discriminative feature learning. (4) Extensive experiments on the newly constructed CEMIRS and CEMIRL datasets demonstrate that SHRC-Net achieves state-of-the-art retrieval accuracy and exhibits strong robustness against ambiguous or adversarial inputs, underscoring its potential for secure deployment in AI-driven consumer electronics systems.

II. RELATED WORK

Unsupervised learning methods train models using only paired data across modalities. Andrew et al. [29] combined canonical correlation analysis with deep neural networks and proposed deep canonical correlation analysis. This method uses DNNs to compute feature mapping matrices for two modalities, which are then projected into a shared feature space to maximize similarity, effectively addressing the challenge of modeling nonlinear relationships that traditional statistical methods struggle with. Wang et al. [30] proposed a modality-specific network design, where each modality is processed by an independent feature extraction network, improving the model's capacity for semantic alignment. Liu et al. [31] introduced an unsupervised cross-modal retrieval method based on generative adversarial networks. This approach uses two feature extractors to map features from different modalities into a shared feature space, while a modality discriminator ensures that unique modality-specific information is preserved during alignment. Lee et al. [32] proposed the stacked cross attention network, a cross-modal retrieval model based on cross-modal attention mechanisms. It aligns features from different modalities through attention-based interaction and trains the model by maximizing the similarity between aligned features. Ji et al. [33] introduced the saliency-guided attention network for cross-modal retrieval, which adopts an asymmetric attention module to learn fine-grained semantic associations across modalities.

Supervised methods mainly leverage labeled data to guide model training. Li et al. [34] proposed a deep neural network-based supervised cross-modal retrieval model, in which separate feature extractors are used for different modalities. The model maximizes the similarity between modality-specific features and label vectors in a shared feature space, enabling fine-grained alignment between modalities. Wei et al. [35] introduced a supervised modality matching method that utilizes pretrained models to extract image features, resulting in more informative image representations. These features are

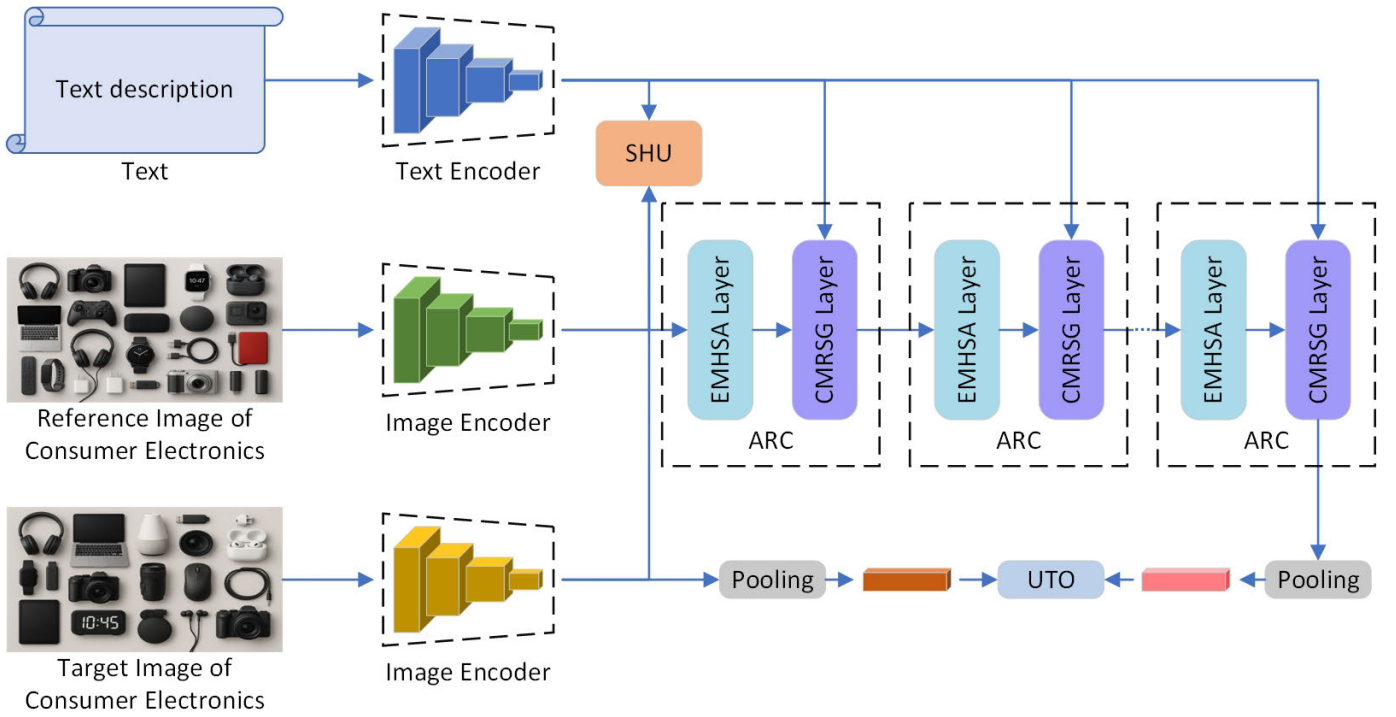


Fig. 1. Structure of the proposed SHRC-Net. It comprises a Semantic Harmonization Unit (SHU) for aligning visual regions with textual semantics, an Adaptive Region Composer (ARC) for modeling spatial and compositional relations under semantic guidance, and a Unified Training Objective (UTO) that jointly optimizes all components via alignment, regularization, and classification losses.

then aligned with label vectors in the common feature space, improving retrieval accuracy. Hu et al. [36] proposed a scalable cross-modal retrieval method that uses separate encoders to extract features from different modalities and decoders to reconstruct them, reducing the information loss introduced during encoding. The model then aligns modality features with label vectors to improve scalability. Xu et al. [37] introduced a contrastive learning approach for cross-modal retrieval that improves robustness to noisy samples. The method applies regularization to denoise feature representations, mitigating the impact of noise during training and enhancing robustness. Zhen et al. [38] proposed deep supervised cross-modal retrieval, which leverages label information to perform fine-grained alignment between different modalities in a shared feature space, ultimately improving retrieval performance.

Self-supervised learning constructs supervision signals from auxiliary tasks to guide semantic alignment between modalities. Liu et al. [31] proposed a self-supervised retrieval method called self-supervised correlation learning, which utilizes semantic similarity between modalities as pseudo-labels to supervise feature alignment, thereby improving model accuracy. Lu et al. [39] introduced a BERT-based pretraining model for cross-modal retrieval. This method employs BERT as a feature extractor for different modalities, first aligning features at a fine-grained level through self-supervised tasks, and then comparing their similarities. Li et al. [40] proposed a multi-modal pretraining model named object-semantics alignment, which inputs both region-level and positional information of images into the model, allowing more accurate cross-modal alignment with the aid of these visual cues. Song

and Soleymani [41] developed a knowledge-enhanced vision-language pretraining model, which constructs scene graphs from the input text to design targeted self-supervised tasks and enhance model performance. Geigle et al. [42] improved upon Oscar by using a parameter-shared Transformer architecture to extract features across modalities, and applied cross-attention for fine-grained semantic alignment, further enhancing retrieval performance.

III. METHOD

A. Overall Framework

This study focuses on multimodal image retrieval for consumer electronics under security-sensitive scenarios and presents a novel cross-modal retrieval framework, SHRC-Net (Semantic Harmonization and Region Composition Network), as illustrated in Fig. 1. The network is designed to handle fine-grained component modifications and user-driven semantic queries while maintaining robustness against adversarial or ambiguous inputs. Given a reference image and a textual query, it effectively interprets user intent and generates a structured representation of the corresponding target image. The architecture consists of three key modules: a Semantic Harmonization Unit (SHU) that aligns visual regions with semantic cues from the query text, an Adaptive Region Composer (ARC) that models spatial and compositional relationships among regions under semantic guidance, and a Unified Training Objective (UTO) that jointly optimizes all modules through alignment, regularization, and classification signals. Together, these components form a coherent pipeline from semantic interpretation to structural feature composition.

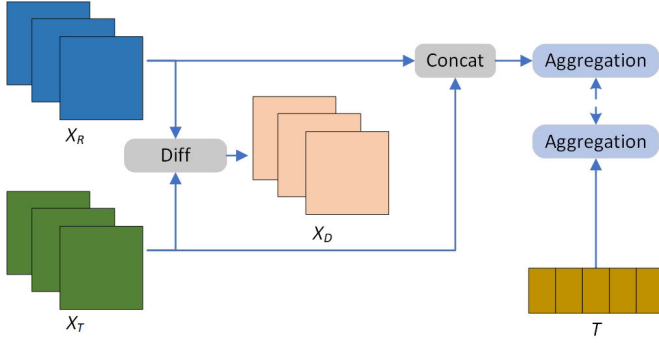


Fig. 2. Structure of the proposed Semantic Harmonization Unit (SHU).

B. Semantic Harmonization Unit (SHU)

In multimodal consumer electronics image retrieval tasks, users typically provide a reference product image and a query text to describe the expected modification of the target image, such as “replace the white back cover” or “remove the fingerprint module.” This type of joint input forms a complex cross-modal semantic matching scenario. One of the core challenges is the modal heterogeneity between images and text: image information often contains fine-grained structural and local details, whereas text encodes high-level semantic abstractions. This is particularly true in the consumer electronics domain, where product images contain many components, have high similarity, and frequent fine structural variations. To ensure semantic consistency across modalities and enhance the model’s ability to capture fine-grained differences, we propose the Semantic Harmonization Unit (SHU), as illustrated in Fig. 2. This module aligns image regions and textual semantics in a shared embedding space, enabling accurate interpretation of diverse user intent. In addition, SHU contributes to the robustness of the overall framework by reinforcing stable region-text associations, even in the presence of ambiguous or adversarial inputs.

This module consists of five stages: basic feature encoding, global alignment, attention-based fusion feature construction, gating mechanism, and regularized representation learning. Each step has a clear function and interacts with adjacent steps via auxiliary mechanisms.

First, during the basic feature encoding stage, the reference image I_r is fed into a visual encoder to extract spatial region-level visual features:

$$V = \{v_1, v_2, \dots, v_n\} \in \mathbb{R}^{n \times d} \quad (1)$$

where each v_i corresponds to a region of the image. For example, in a smartphone image, v_i may represent the screen, edge, or camera area. To preserve region completeness and semantic consistency, we adopt a uniform intermediate-level segmentation strategy to ensure fine-grained coverage.

Simultaneously, the query text T is encoded using a language encoder to produce word vectors:

$$W = \{w_1, w_2, \dots, w_m\} \in \mathbb{R}^{m \times d} \quad (2)$$

where each w_j represents the contextual representation of the j -th word in the sentence. Although query text is short,

it contains explicit modification intent making accurate region-word alignment essential.

To evaluate whether the image and text share consistent global semantics, we first perform average pooling on both modalities. The pooled region feature is denoted $f_{\text{img}}^{\text{pool}}(V)$ and $f_{\text{text}}^{\text{pool}}(W)$, the global alignment score is computed as:

$$s(I_r, T) = \frac{(f_{\text{img}}^{\text{pool}}(V))^{\top} f_{\text{text}}^{\text{pool}}(W)}{\|f_{\text{img}}^{\text{pool}}(V)\| \cdot \|f_{\text{text}}^{\text{pool}}(W)\|} \quad (3)$$

This score reflects the global semantic consistency between the image and the text, especially for descriptions of overall changes. It serves as a weak supervision signal during the early matching stage.

However, global similarity is less effective in capturing partial semantic transformations. Text usually involves local changes in the image, so we introduce an attention mechanism to establish interactions between image regions and text tokens. The similarity between each image region v_i and text word w_j is computed as:

$$\alpha_{ij} = \frac{\exp(v_i^{\top} w_j)}{\sum_{k=1}^m \exp(v_i^{\top} w_k)} \quad (4)$$

This matrix acts as the influence score of text tokens on image regions, helping build soft semantic alignments between the two. In this attention mechanism, we compute weighted aggregated image features as:

$$\tilde{v} = \sum_{i=1}^n \sum_{j=1}^m \alpha_{ij} v_i, \quad \tilde{w} = \sum_{j=1}^m w_j \quad (5)$$

Here, $\tilde{v}$ represents the focused image representation guided by text semantics, and $\tilde{w}$ is the aggregated query embedding. Both are projected into a shared space for contrastive learning to achieve precise alignment.

For contrastive learning, we adopt a bidirectional strategy. The image-to-text loss is defined as:

$$\mathcal{L}_{i2t} = -\log \frac{\exp(\text{sim}(\tilde{v}, \tilde{w})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\tilde{v}, \tilde{w}_j)/\tau)} \quad (6)$$

The text-to-image loss is:

$$\mathcal{L}_{t2i} = -\log \frac{\exp(\text{sim}(\tilde{w}, \tilde{v})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\tilde{w}, \tilde{v}_j)/\tau)} \quad (7)$$

Taking the average of both directions, the final alignment loss becomes:

$$\mathcal{L}_{\text{align}} = \frac{1}{2B} \sum_{i=1}^B (\mathcal{L}_{i2t}^{(i)} + \mathcal{L}_{t2i}^{(i)}) \quad (8)$$

This mechanism enables hard negative sample construction within the batch, allowing the model to distinguish subtle semantic differences across similar samples.

Considering that the model may still be disturbed by irrelevant regions, we further introduce a semantic gating mechanism (Semantic Gating), which generates global gating vectors from the pooled features:

$$g_v = \sigma(W_v^{\top} f_{\text{img}}^{\text{pool}}(V)), \quad g_w = \sigma(W_w^{\top} f_{\text{text}}^{\text{pool}}(W)) \quad (9)$$

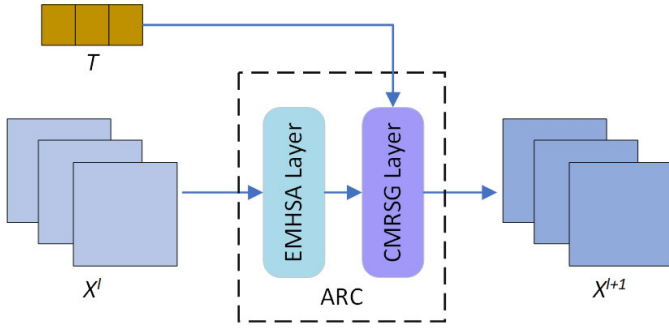


Fig. 3. Structure of the proposed Semantic Harmonization Unit (SHU).

The gating vectors are then applied to the region features as:

$$v'_i = g_v \odot v_i, \quad w'_j = g_w \odot w_j \quad (10)$$

This operation can be viewed as channel-wise filtering—suppressing irrelevant information, and enhancing semantics-related features such as color, texture, and shape. It is especially applicable in scenes with complex backgrounds or varying viewing angles.

To prevent sparse matching or unbalanced attention distribution, we design the following regularization term:

$$\mathcal{L}_{\text{reg}} = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m (\text{sim}(v'_i, w'_j) - \bar{s})^2 \quad (11)$$

Here, $\bar{s}$ is the average similarity across all region-word pairs. This regularization encourages the similarity distribution to be more stable and concentrated, reducing gradient variance during training.

The final objective function is:

$$\mathcal{L}_{\text{SHU}} = \mathcal{L}_{\text{align}} + \beta \cdot \mathcal{L}_{\text{reg}} \quad (12)$$

C. Adaptive Region Composer (ARC)

After cross-modal alignment, the system must further integrate regional visual features and semantic information to construct a unified representation that reflects the user's fine-grained retrieval intent. In multimodal consumer electronics image retrieval tasks, this step is particularly crucial, as user-desired changes often affect specific local regions, such as modifying the camera module or adjusting the bezel thickness. Failure to capture these granular semantic-to-visual relationships can lead to inaccurate retrievals. To address this, we propose the Adaptive Region Composer (ARC), as illustrated in Fig. 3, which builds on the output of SHU and incorporates a refined self-attention mechanism to strengthen intra-region relationships and enhance the alignment of visual and textual representations at the region level. Furthermore, ARC contributes to the robustness of the system by facilitating stable region composition, even when facing structurally ambiguous or adversarial inputs.

The ARC module consists of two main components: one is an Enhanced Multi-Head Self-Attention (EMHSA) mechanism to capture contextual dependencies among image regions; the other is a Cross-Modal Region-Semantic Guidance (CMRSG) module to emphasize query-relevant semantic cues. We denote

the regional feature map as $X^l \in \mathbb{R}^{K \times C}$, where K is the number of regions, and C is the channel dimension. This representation encodes spatial structure and local features and is the basis for region composition.

In the enhanced attention module, we adopt the standard Transformer attention as backbone and introduce a Region-Gated Attention Mask (RGAM) and a Local Denoising Term (LDT) to improve focus and suppress noise. These mechanisms are specifically designed for consumer electronics scenarios where image backgrounds are complex and often contain irrelevant information like environments, logos, and auxiliary objects. For region sequence X^l , we first compute the queries, keys, and values:

$$Q = X^l W_Q, \quad K = X^l W_K, \quad V = X^l W_V \quad (13)$$

We then inject region relevance gating into attention weights:

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{\text{gate}}\right) V \quad (14)$$

Here,

$$M_{\text{gate}}(i, j) = \sigma\left(w_g^\top [x_i; x_j]\right) \cdot \delta_{ij} \quad (15)$$

where δ_{ij} is the distance-based adjacency mask between regions, and σ is the Sigmoid activation. This gating mask adjusts attention based on relative distance and semantic correlation, suppressing noisy region interactions. For instance, when describing “move the camera upward”, this mechanism strengthens attention to structurally relevant regions such as camera, bezel, and logo.

Additionally, dynamic masking is employed to improve robustness against structural variation. In consumer electronics, adjacent regions may vary greatly in edge thickness or positioning, requiring selective attention. The updated attention output is:

$$\hat{X}^l = \text{LayerNorm}(\text{MultiHead}(X^l) + X^l) \quad (16)$$

To prevent attention from collapsing on redundant regions, we introduce a denoising loss:

$$\mathcal{L}_{\text{denoise}} = \sum_{i=1}^K \left(1 - \frac{1}{H} \sum_{h=1}^H \alpha_i^h\right)^2 \quad (17)$$

Here, α_i^h denotes the attention weight assigned to region i in head h . This term encourages diverse region attention and prevents domination by a few large areas. It is especially effective for retrieval involving minor changes like subtle design shifts.

After context encoding, we inject textual guidance to enrich visual representations. Given text tokens $T = \{t_1, t_2, \dots, t_M\} \in \mathbb{R}^{M \times C}$, we compute cross-modal attention between each region feature x_i^l and text token t_j :

$$A_{ij} = \text{Softmax}_j \left(\frac{(x_i^l W_Q^t)(t_j W_K^t)^\top}{\sqrt{d_k}} \right) \quad (18)$$

$$\hat{x}_i^l = \sum_{j=1}^M A_{ij} (t_j W_V^t) \quad (19)$$

This results in a cross-modal semantic attention map, allowing each image region to incorporate weighted contextual semantics from query text. It supports complex operations such as “move fingerprint sensor upward,” enabling fine-grained feature adjustment. We concatenate original and guided features for fusion.

To enhance interpretability and control, we also introduce a saliency-aware gating mechanism. When a text token explicitly targets a region, attention is focused on matching target regions. This balances model attention and prevents semantic drift. To reduce interference from similar but irrelevant regions, the model learns to isolate inconsistent query-visual interactions.

The final fusion is processed through a residual feedforward network:

$$\hat{x}_i = \text{ReLU}(W_1[x_i^l; \hat{x}_i^l] + b_1) \quad (20)$$

This is followed by a residual layer:

$$X^{l+1} = \text{LayerNorm}(\hat{X} + X^l) \quad (21)$$

Through the above process, ARC adaptively composes visual representations under text guidance. It aggregates spatial context and semantic relevance to support retrieval-level understanding. Compared with previous methods that use static fusion, ARC enables dynamic modulation of regional features under weak supervision, offering better generalization and compatibility with SHU.

D. Unified Training Objective (UTO)

To achieve accurate retrieval in multimodal scenarios within consumer electronics, we design a unified training objective composed of multiple loss components. This framework jointly optimizes cross-modal alignment, region-level attention regularization, and semantic classification. By integrating these objectives, the network learns to produce structured, interpretable, and semantically aligned feature representations.

The semantic alignment loss $\mathcal{L}_{\text{SHU}}$ supervises region-to-text alignment and distributional consistency during the early stages of training.

To ensure stable and diverse region-level attention, we introduce an attention regularization loss $\mathcal{L}_{\text{attn}}$. This term penalizes overly concentrated attention distributions across multiple heads and promotes richer, more distributed structural representations.

Given H attention heads and $K \times K$ attention maps $A^h \in \mathbb{R}^{K \times K}$, the entropy of each head is computed as:

$$H(A_i^h) = - \sum_{j=1}^K A_{ij}^h \log A_{ij}^h \quad (22)$$

The average entropy across all heads and positions defines the attention regularization loss:

$$\mathcal{L}_{\text{attn}} = - \frac{1}{H \cdot K} \sum_{h=1}^H \sum_{i=1}^K H(A_i^h) \quad (23)$$

This encourages more uniform and interpretable attention distributions, avoiding excessive activation on a few dominant regions and enhancing model generalizability.

To guide the final retrieval decision, we incorporate a classification loss $\mathcal{L}_{\text{cls}}$ based on the similarity between the composed feature and the target image representation:

$$\mathcal{L}_{\text{cls}} = - \frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(o_i, x_{t_i}))}{\sum_{j=1}^B \exp(s(o_i, x_{t_j}))} \quad (24)$$

where B is the batch size, and $s(\cdot, \cdot)$ is the similarity function between the composed output o_i and the ground truth image feature x_{t_i} . This loss encourages discriminative target matching and improves the model's decision confidence.

The overall training objective combines all three components:

$$\mathcal{L}_{\text{total}} = \lambda_1 \cdot \mathcal{L}_{\text{SHU}} + \lambda_2 \cdot \mathcal{L}_{\text{attn}} + \lambda_3 \cdot \mathcal{L}_{\text{cls}} \quad (25)$$

This weighting ensures that early training emphasizes cross-modal alignment, while later stages focus on attention stability and semantic discrimination. The overall configuration enables robust retrieval performance in fine-grained, multi-component scenarios typical in consumer electronics applications.

IV. EXPERIMENT

A. Dataset and Performance Metric

To evaluate the effectiveness and robustness of our proposed method in multimodal image retrieval scenarios specific to consumer electronics, we construct two custom datasets: CEMIRS and CEMIRL. CEMIRS is a small-scale dataset containing 46,053 training samples and 17,551 test samples. It includes common consumer electronics categories such as smartphones, laptops, and smartwatches. Each sample consists of a reference image, a textual modification query, and the corresponding target image. The dataset focuses on fine-grained component-level changes under moderate variability. CEMIRL extends CEMIRS with more categories and more complex variations, including camera adjustments, frame changes, and component shifts. It comprises 128,956 training samples and 38,942 test samples, providing a more challenging benchmark with diverse product structures and modification types. Together, these datasets enable comprehensive evaluation under both typical and adversarial-like conditions, offering a basis for assessing the retrieval system's accuracy, structural sensitivity, and resilience to ambiguous or potentially manipulated inputs. For evaluation, we adopt the widely used Recall@k metric, which measures the percentage of queries for which the correct target image is ranked within the top-k results.

B. Experimental Setup

In all experiments, we implement our model using PyTorch and conduct training and evaluation on a single NVIDIA RTX 3090 GPU. The proposed network is trained in an end-to-end fashion using the Adam optimizer with a batch size of 64. The learning rate is initialized to 1×10^{-4} and decayed linearly throughout training. The model is trained for 50 epochs on each dataset, and the best model is selected based on validation Recall@5 performance.

TABLE I

COMPARISON OF RETRIEVAL PERFORMANCE ON CEMIRS AND CEMIRL

Method	CEMIRS		CEMIRL	
	Recall@1	Recall@5	Recall@1	Recall@5
RELA [43]	61.2	84.7	63.8	86.9
FiLM [44]	65.4	88.2	68.1	89.5
CAE [45]	68.7	90.1	71.6	91.8
VAL [46]	71.3	91.7	74.5	93.2
CLRPLANT [47]	73.6	93.1	77.2	94.5
CosMo [48]	74.5	94.0	78.5	95.0
LEmpo [49]	75.5	94.2	79.1	95.6
Ours	79.3	95.6	82.9	96.9

For visual feature extraction, we use a Vision Transformer (ViT) as the image encoder. Specifically, we adopt a pretrained ViT-base model to extract patch-level features from the reference and target images. The extracted region features are then fed into the SHU and ARC modules for semantic alignment and compositional reasoning. For textual encoding, we employ a bidirectional GRU (BiGRU) to encode the query text. Each word is embedded using a pretrained word embedding layer, and the BiGRU outputs a sequence of contextualized token features. These textual features are used in both the SHU module for alignment and the ARC module for cross-modal interaction.

C. Comparison With Existing Methods

To verify the effectiveness of the proposed model, we compare it against six prior approaches referenced from existing literature. These methods, corresponding to RELA [43], FiLM [44], CAE [45], VAL [46], CLRPLANT [47], CosMo [48], and LEmpo [49], represent various paradigms in cross-modal and multimodal retrieval. All models are re-trained and evaluated under consistent experimental settings on our datasets for fair comparison. Table I presents the results on the CEMIRS and CEMIRL datasets, evaluated using Recall@1 and Recall@5. For each metric, our proposed method achieves the highest performance across both datasets. As shown in Table I, our model achieves the best performance among all methods. On the CEMIRS dataset, our approach reaches 79.3% Recall@1 and 95.6% Recall@5, outperforming the second-best result by 3.8% and 1.4% respectively. On the larger and more challenging CEMIRL dataset, our model achieves 82.9% Recall@1 and 96.9% Recall@5, maintaining consistent superiority. The improvements validate the effectiveness of the proposed semantic alignment and region composition strategy, which allows our model to better capture user-intended modifications in both visual and structural aspects. While all other methods improve from CEMIRS to CEMIRL, likely due to increased data volume and diversity, our method exhibits the most robust performance across both scales.

D. Ablation Study on SHU and ARC Modules

To further validate the effectiveness of the proposed Semantic Harmonization Unit (SHU) and Adaptive Region Composer (ARC), we conduct detailed ablation studies by

TABLE II

ABLATION RESULTS ON CEMIRS AND CEMIRL

Model	CEMIRS		CEMIRL	
	R@1	R@5	R@1	R@5
Full SHRC-Net (SHU + ARC)	79.3	95.6	82.9	96.9
w/o ARC (only SHU)	75.8	93.4	79.2	94.6
w/o SHU (only ARC)	74.1	92.7	77.9	94.0
w/o SHU and ARC	70.4	90.2	74.8	91.3

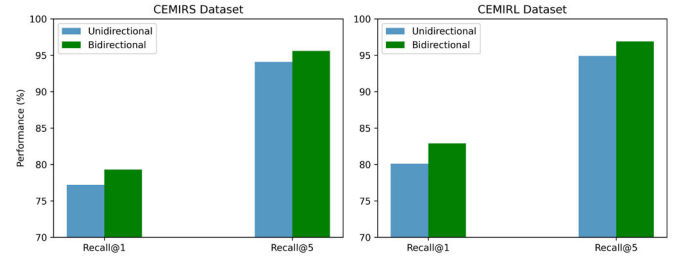


Fig. 4. Analysis of bidirectional contrastive learning in SHU.

selectively disabling one or both modules from the full SHRC-Net framework. Table II reports the retrieval performance in terms of Recall@1 and Recall@5 on both CEMIRS and CEMIRL datasets. As shown in Table II, removing either SHU or ARC results in a noticeable drop in performance. The model without ARC drops 3.5% on Recall@1 in CEMIRS, confirming the importance of region composition in structuring visual features under semantic guidance. Similarly, excluding SHU causes a 5.2% drop on CEMIRL in Recall@1, demonstrating that fine-grained cross-modal alignment is critical in capturing the user's intent. The performance degradation becomes even more pronounced when both SHU and ARC are removed, especially in complex product variation scenarios in CEMIRL. This indicates that these modules are not only individually beneficial but also mutually reinforcing. SHU enhances semantic interpretability by grounding text in specific regions, while ARC ensures that these regions interact in a structurally coherent manner. Without their combined effect, the model tends to rely on superficial feature fusion, limiting its ability to recognize subtle changes. These results quantitatively confirm that the synergy of SHU and ARC is essential to the overall effectiveness of SHRC-Net, particularly for retrieval tasks involving highly structured objects and fine-grained modifications common in consumer electronics.

E. Effect of Bidirectional Contrastive Learning in SHU

To investigate the effectiveness of the bidirectional contrastive learning strategy employed in the SHU module, we conduct controlled experiments comparing it against a unidirectional variant. Specifically, the unidirectional model computes contrastive loss only from image to text (i2t), while our proposed version symmetrically includes both image-to-text (i2t) and text-to-image (t2i) objectives. Fig. 4 illustrates the retrieval performance in terms of Recall@1 and Recall@5 under both strategies on the CEMIRS and CEMIRL datasets. The bidirectional contrastive learning strategy consistently outperforms the unidirectional variant across both datasets. On the CEMIRS dataset, it improves Recall@1 by 2.1%

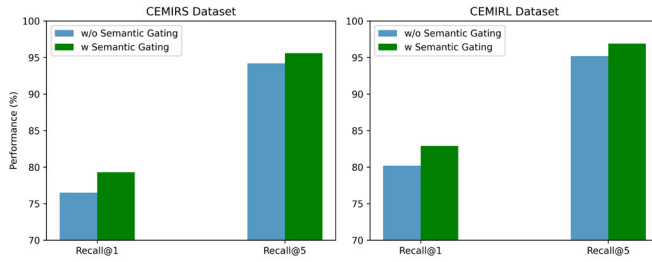


Fig. 5. Analysis of semantic gating mechanism in SHU.

and Recall@5 by 1.5%. The margin is even larger on the more complex CEMIRL dataset, with gains of 2.8% on Recall@1 and 2.0% on Recall@5. These results indicate that incorporating bidirectional matching strengthens semantic alignment and reduces ambiguity in image-text associations. The improvement is particularly evident in cases involving subtle query modifications, where single-direction alignment may ignore critical contextual dependencies. Bidirectional supervision helps the model establish more robust feature correspondences from both perspectives, thereby enhancing the discriminative power of the learned embeddings. This dual-view consistency is essential for capturing nuanced user intent in fine-grained retrieval scenarios.

F. Effect of the Semantic Gating Mechanism in SHU

To assess the contribution of the semantic gating mechanism in the SHU module, we conduct experiments by disabling this component while keeping all other configurations unchanged. This mechanism is designed to filter out irrelevant visual or textual features by applying learned gating vectors on both region-level image features and word-level text embeddings. It helps suppress background noise and enhance semantic saliency, especially in cluttered product images or when the query describes local changes. Fig. 5 presents the retrieval performance on the CEMIRS and CEMIRL datasets under two settings: with and without the semantic gating mechanism. As shown in the figure, disabling the gating mechanism leads to a consistent drop in performance across both datasets. On CEMIRS, Recall@1 declines from 79.3% to 76.5%, while Recall@5 drops by 1.4%. On the more challenging CEMIRL dataset, the performance degradation is more pronounced, with Recall@1 reduced by 2.7%. This result suggests that the gating mechanism plays a key role in improving feature discriminability by reinforcing relevant channels and diminishing noisy or redundant information. In qualitative retrieval cases, models without gating tend to highlight broader or ambiguous regions, often confusing similar-looking components. With gating enabled, attention distributions become more focused on semantically relevant parts, leading to more accurate region-word alignment and ultimately better retrieval quality. Therefore, the semantic gating mechanism proves to be an effective enhancement in SHU for fine-grained multimodal understanding.

G. Effect of the Enhanced Self-Attention Mechanism in ARC

The Adaptive Region Composer (ARC) module in SHRC-Net incorporates an enhanced multi-head self-attention

TABLE III
PERFORMANCE COMPARISON OF STANDARD AND ENHANCED ATTENTION MECHANISMS IN ARC

Attention Type	CEMIRS		CEMIRL	
	Recall@1	Recall@5	Recall@1	Recall@5
Standard MHSA	77.5	94.0	79.6	95.0
Enhanced MHSA	79.3	95.6	82.9	96.9

TABLE IV
PERFORMANCE COMPARISON OF RGAM AND LDT IN ARC ON CEMIRS AND CEMIRL DATASETS

ARC Configuration	CEMIRS		CEMIRL	
	Recall@1	Recall@5	Recall@1	Recall@5
w/o RGAM/LDT	76.2	94.1	78.5	95.2
w/ RGAM	77.4	94.8	80.3	95.9
w/ LDT	77.0	94.5	80.1	95.7
w/ RGAM/LDT	79.3	95.6	82.9	96.9

mechanism that differs from the standard Transformer-style attention. To evaluate the effectiveness of these modifications, we replace the enhanced attention mechanism in ARC with a standard multi-head self-attention block and compare performance under otherwise identical settings. Table III summarizes the results on both CEMIRS and CEMIRL datasets. Replacing the standard self-attention with our enhanced design consistently improves performance across both datasets. On CEMIRL, which involves more diverse and subtle structural variations, the improvement reaches 3.3% on Recall@1 and 1.9% on Recall@5, indicating that the enhanced attention enables more precise modeling of spatial dependencies and semantic relevance. These results demonstrate the benefit of incorporating inductive biases such as regional gating and attention denoising into the ARC module, particularly in product-level retrieval scenarios that require detailed structural reasoning.

H. Ablation Study on RGAM and LDT in ARC

To further analyze the contribution of each enhancement in the Adaptive Region Composer (ARC), we conduct an ablation study focusing on the Region-Gated Attention Mask (RGAM) and the Local Denoising Term (LDT). RGAM is designed to refine attention weights by emphasizing spatially and semantically relevant region pairs, while LDT aims to prevent attention collapse and ensure diversity across attention heads. Both components are intended to improve the compositional reasoning capability of ARC in fine-grained product retrieval scenarios. We compare four configurations: the ARC module without either mechanism, with only RGAM, with only LDT, and with both RGAM and LDT (i.e., the full version of ARC). Table IV reports the performance under each setting. Both RGAM and LDT individually contribute to performance improvements over the base ARC configuration. RGAM has a slightly stronger impact than LDT, likely due to its ability to filter irrelevant spatial dependencies during attention computation. However, the best results are obtained when both mechanisms are applied jointly. This synergy indicates that RGAM and LDT address complementary challenges: RGAM focuses on the relevance of attention connections, while LDT

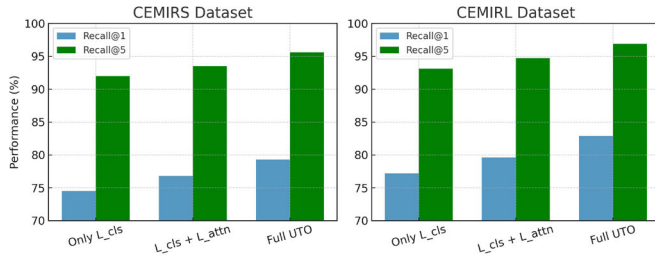


Fig. 6. Analysis of multi-loss design in UTO.

improves attention robustness and distribution. Their combination leads to the most accurate region-level reasoning and compositional feature synthesis, especially evident in the structurally complex CEMIRL dataset.

I. Incremental Analysis of Multi-Loss Design in UTO

The Unified Training Objective (UTO) in SHRC-Net jointly optimizes three complementary loss functions: (1) classification loss L_{cls} to enforce instance-level discriminability, (2) attention regularization loss L_{attn} to encourage stable and diverse attention patterns, and (3) semantic alignment loss L_{SHU} to enhance region-text correspondence. To evaluate the individual and combined impact of these losses, we perform an incremental study by progressively adding each term to the training process. Fig. 6 reports Recall@1 and Recall@5 on the CEMIRS and CEMIRL datasets under three settings: using only L_{cls} , using $L_{cls} + L_{attn}$, and using the full UTO loss. The results demonstrate that each added loss term yields consistent gains. On CEMIRS, Recall@1 increases from 74.5% to 79.3% and Recall@5 from 92.0% to 95.6% when all three loss terms are included. On CEMIRL, the performance improves from 77.2% to 82.9% (Recall@1) and from 93.1% to 96.9% (Recall@5). The attention regularization helps the model capture more reliable structural cues, while the semantic alignment term enhances the correspondence between visual regions and textual instructions. These results validate the design of UTO as an effective multi-loss objective for learning robust and semantically grounded representations.

J. Effect of Batch Size on Retrieval Performance

To investigate the impact of batch size on retrieval performance, we conduct a controlled experiment by varying the batch size from 16 to 256 in powers of two: {16, 32, 64, 96, 128, 256}. For each configuration, the model is trained and evaluated under identical settings, and we report Recall@1 and Recall@5 on both CEMIRS and CEMIRL datasets. Retrieval performance improves as the batch size increases from 16 to 64, reaching a maximum at 64. Beyond this point, further increasing the batch size leads to a decline in performance. This trend is observed consistently across both Recall@1 and Recall@5 metrics for both datasets. The improvement with small to moderate batch sizes can be attributed to better negative sample diversity during contrastive learning, which enhances the model's ability to discriminate fine-grained semantic differences.

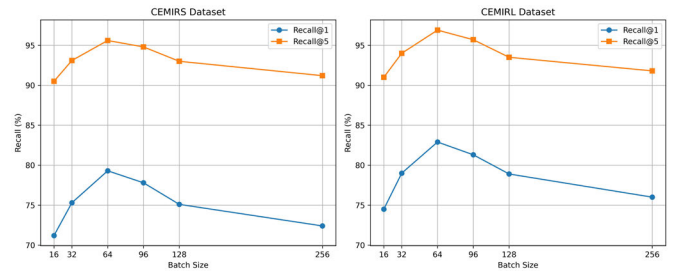


Fig. 7. Analysis of batch size on retrieval performance.

V. CONCLUSION

In this work, we proposed SHRC-Net, a novel multimodal image retrieval framework designed for consumer electronics scenarios, where users often express retrieval intent through a combination of reference images and textual modifications. SHRC-Net integrates three key components: the Semantic Harmonization Unit (SHU) for fine-grained region-text alignment, the Adaptive Region Composer (ARC) for structure-aware feature composition, and the Unified Training Objective (UTO) for robust and discriminative representation learning. Extensive experiments on benchmark datasets demonstrate that SHRC-Net achieves state-of-the-art performance in capturing detailed user intent and retrieving products with subtle structural and semantic variations.

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